Practical No – 1

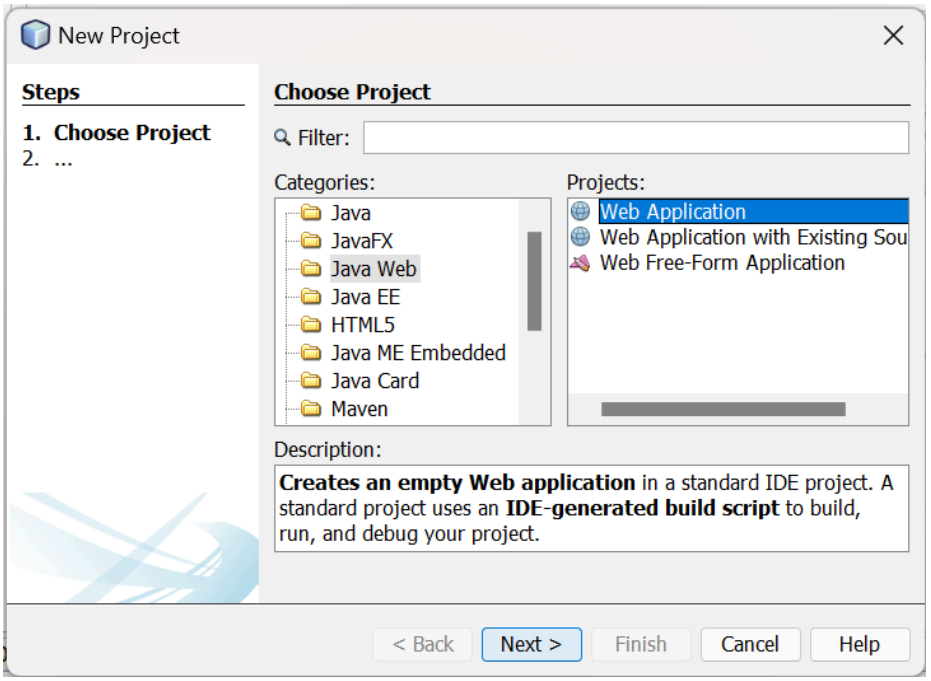
Aim: Write a program to create a simple web service that performs arithmetic operations.

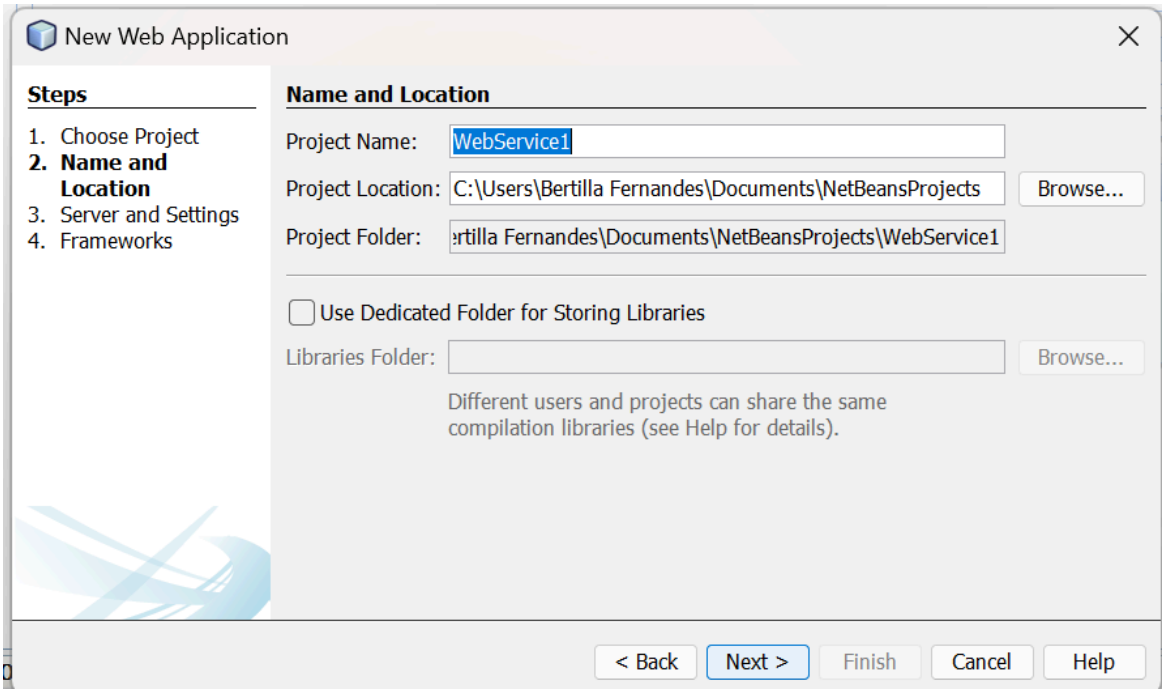
Code:

Steps:

A. Create the WebService:

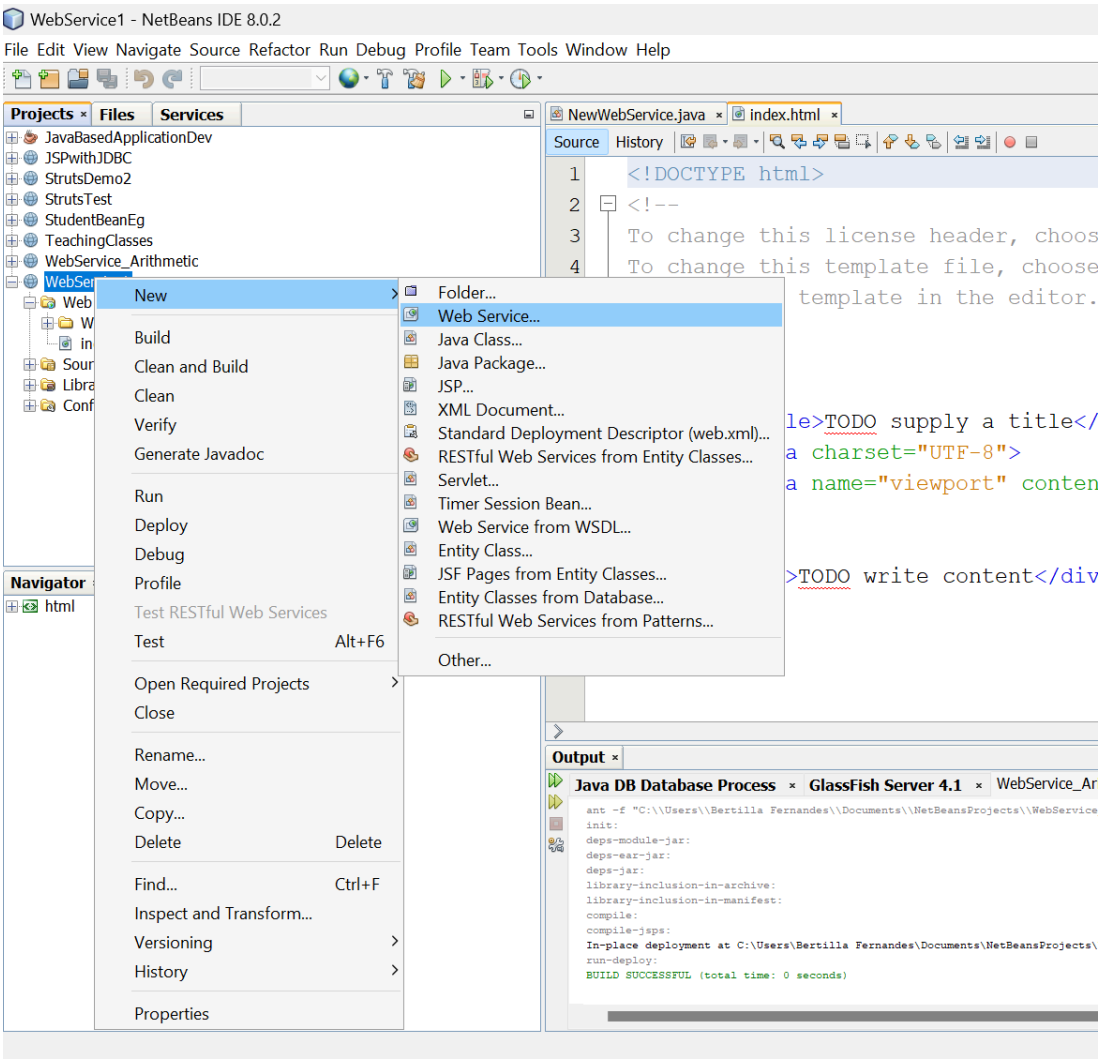
1. Go to File -> New Project. Select Java Web in categories and Web Application in Projects. Click on Next to create web based project.



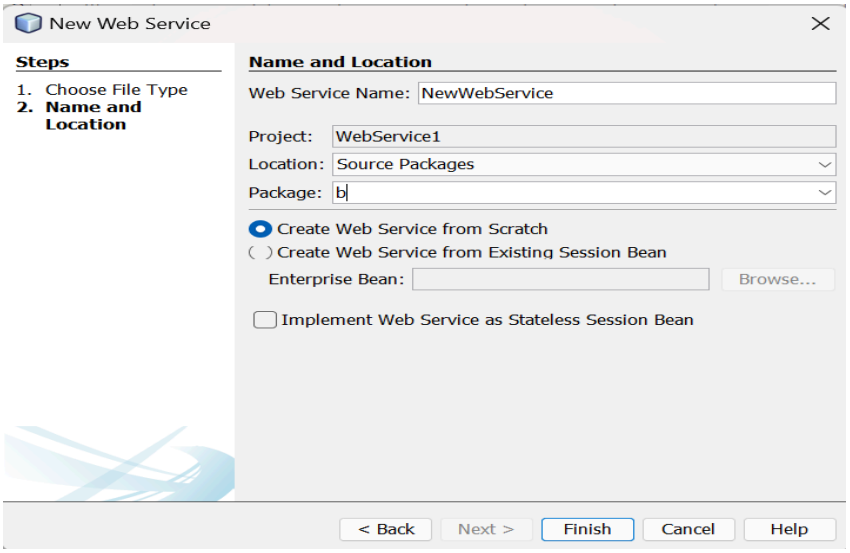


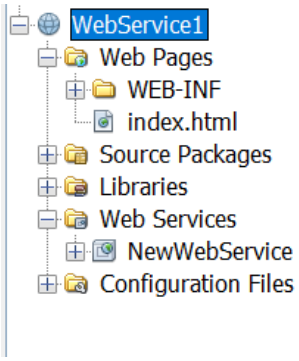
Click on Next and Finish.

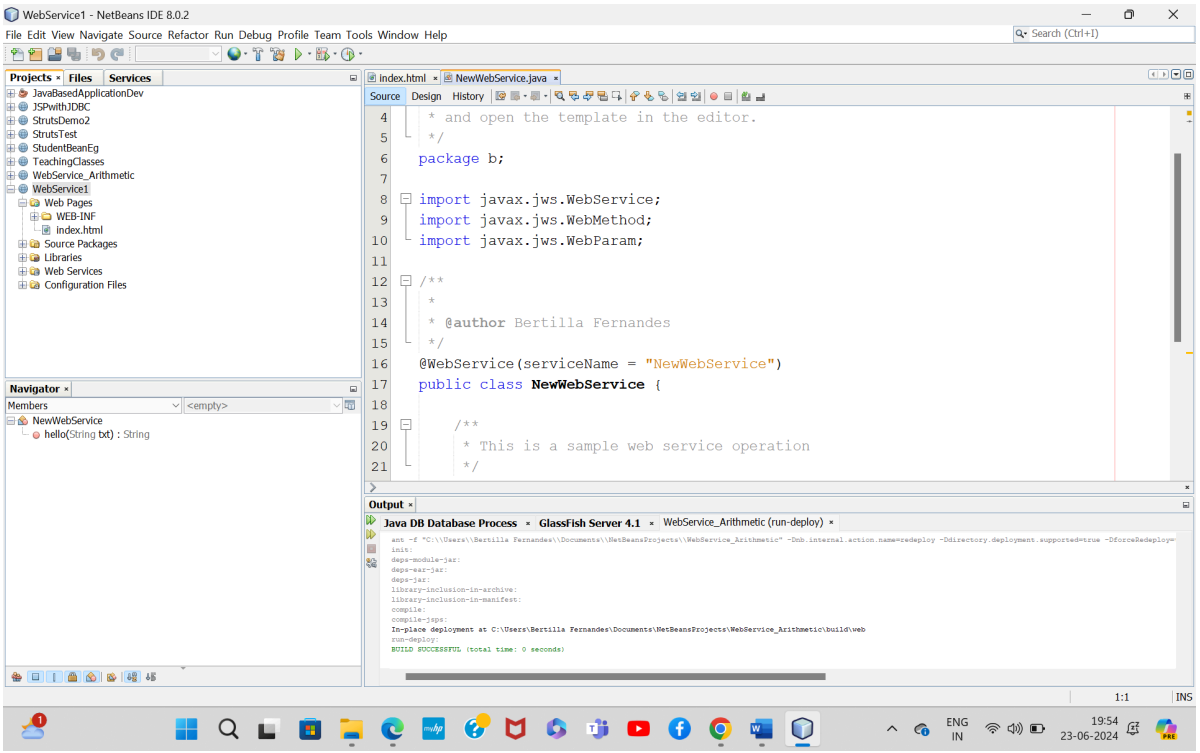
2. Create web service. Right click on Project -> New -> Web Service



3. Enter a Web Service Name and package name and then click on Finish to create a Web Service.



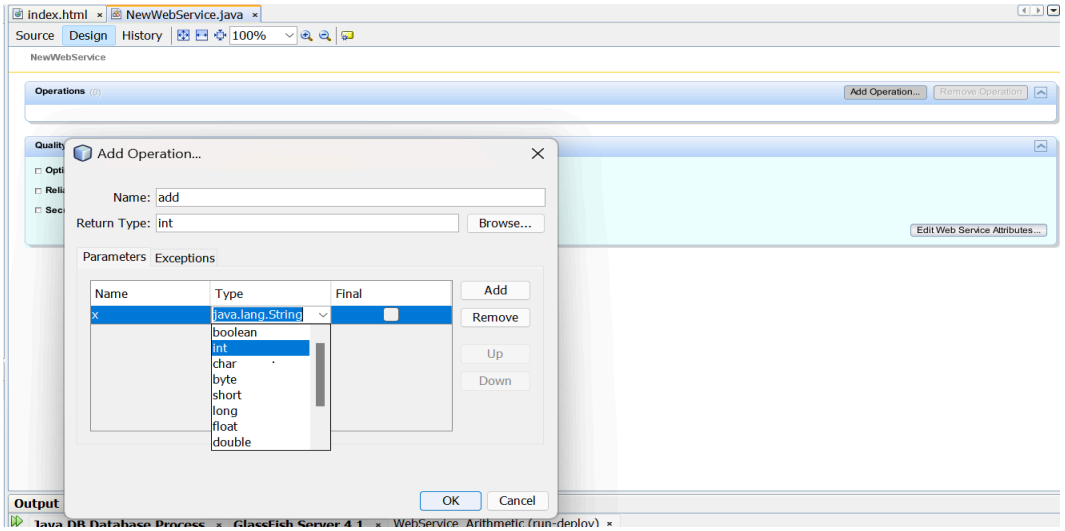


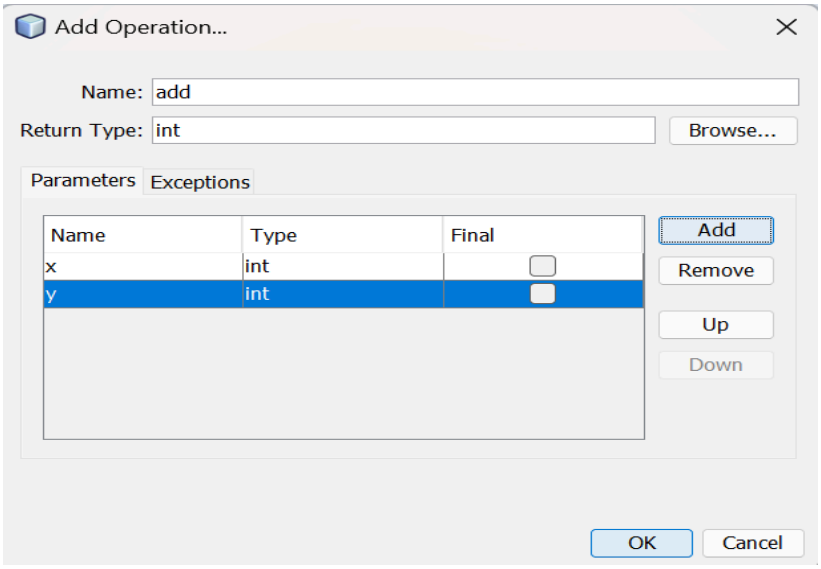


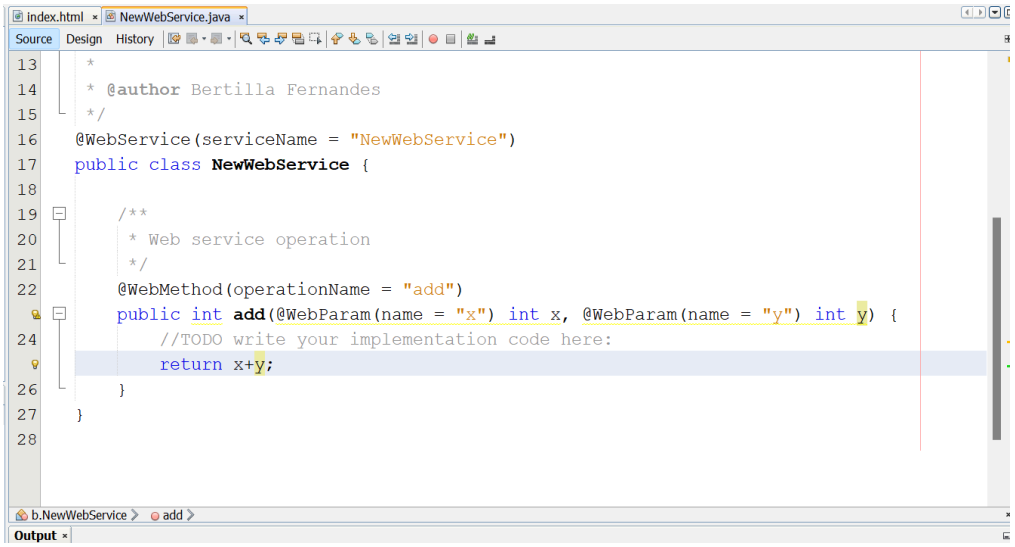
4. Click on Design Tab:

Click on hello and then click on “Remove Operation”.

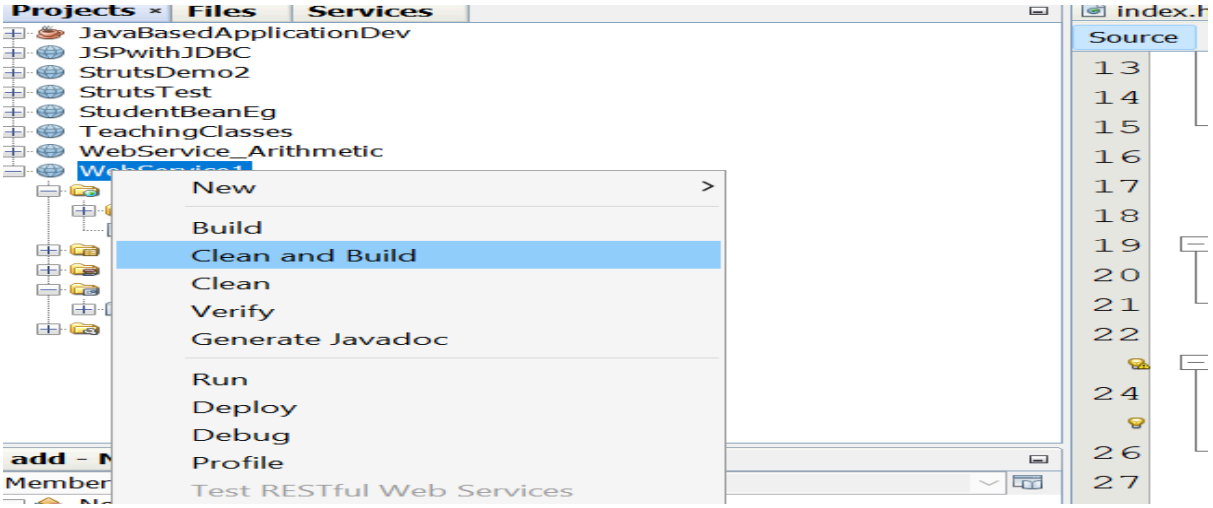
Now Click on “Add Operation” to add operation.

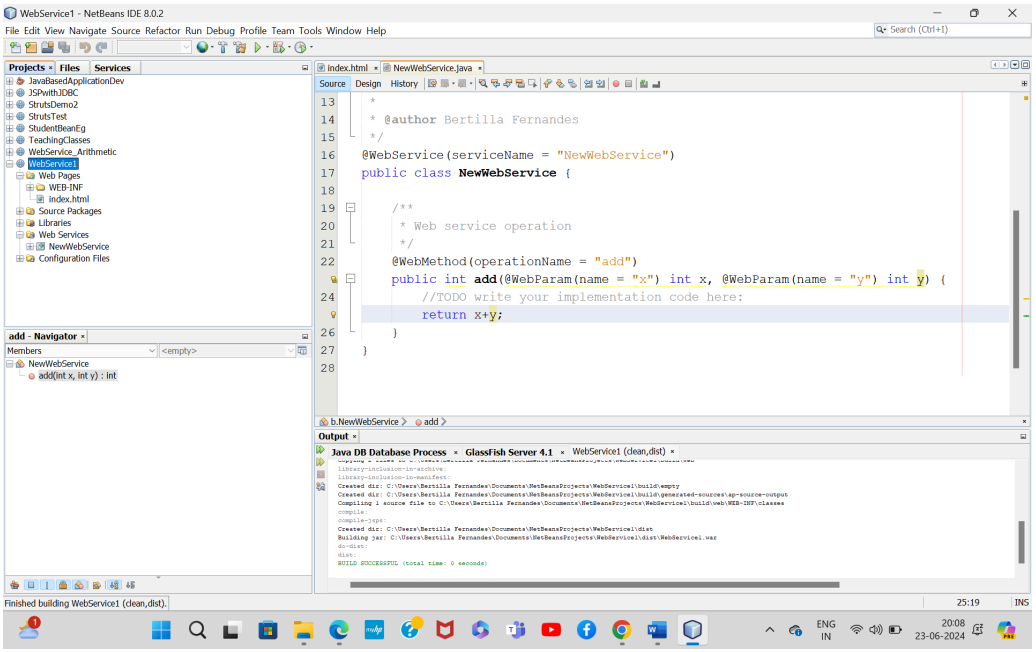




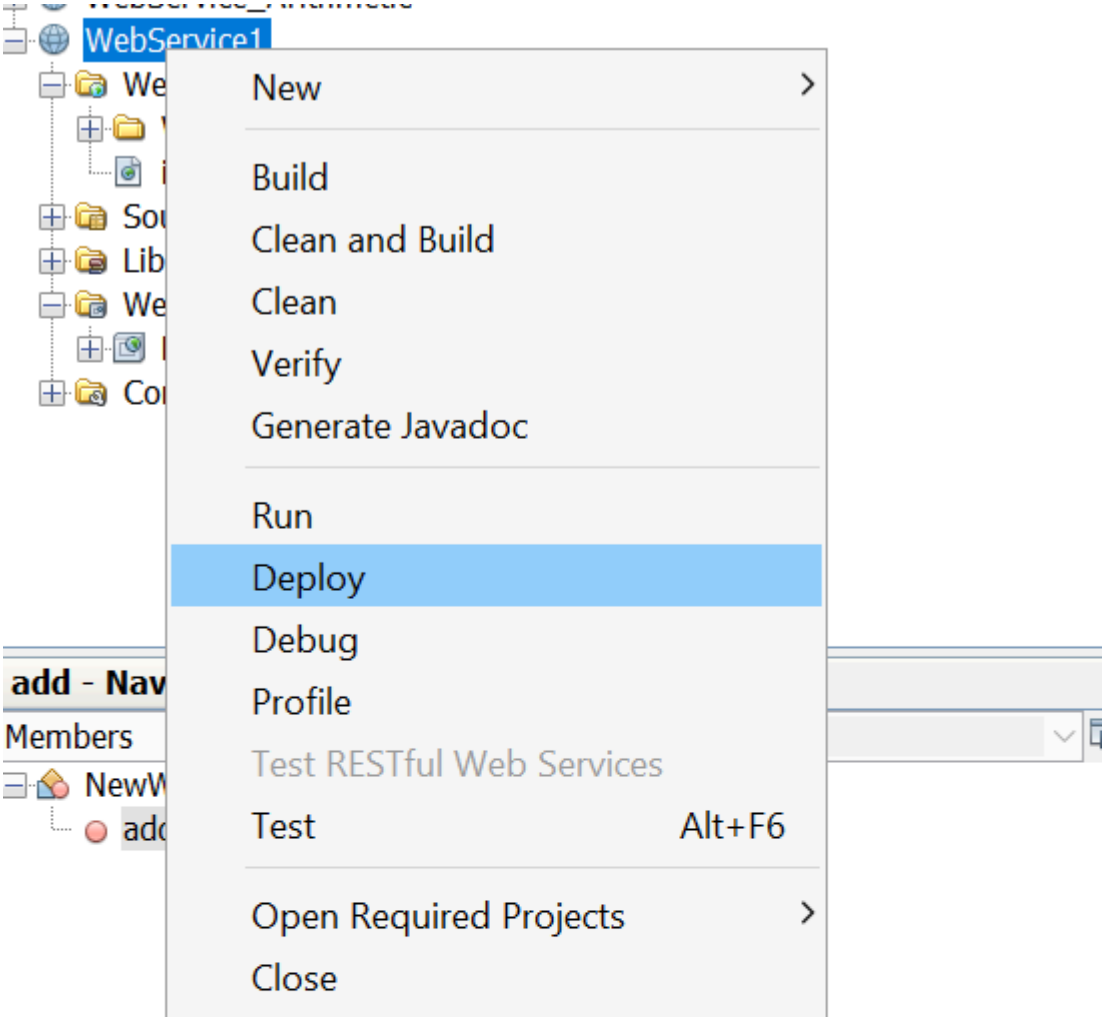


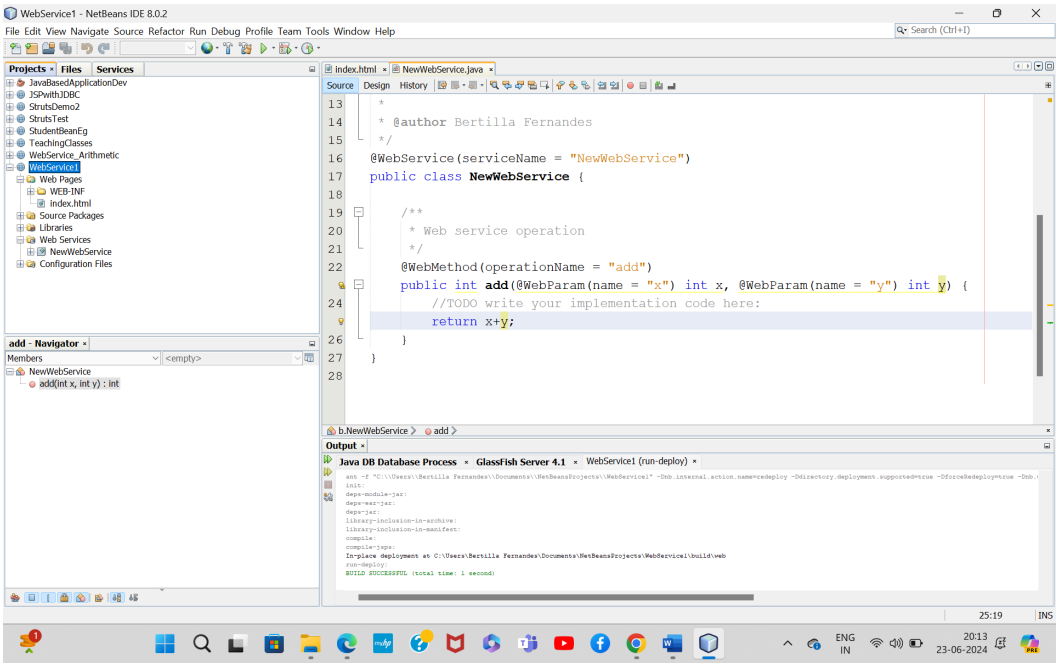
5. Now right click on project name and click on “Clean and Build” to clean and build your project.



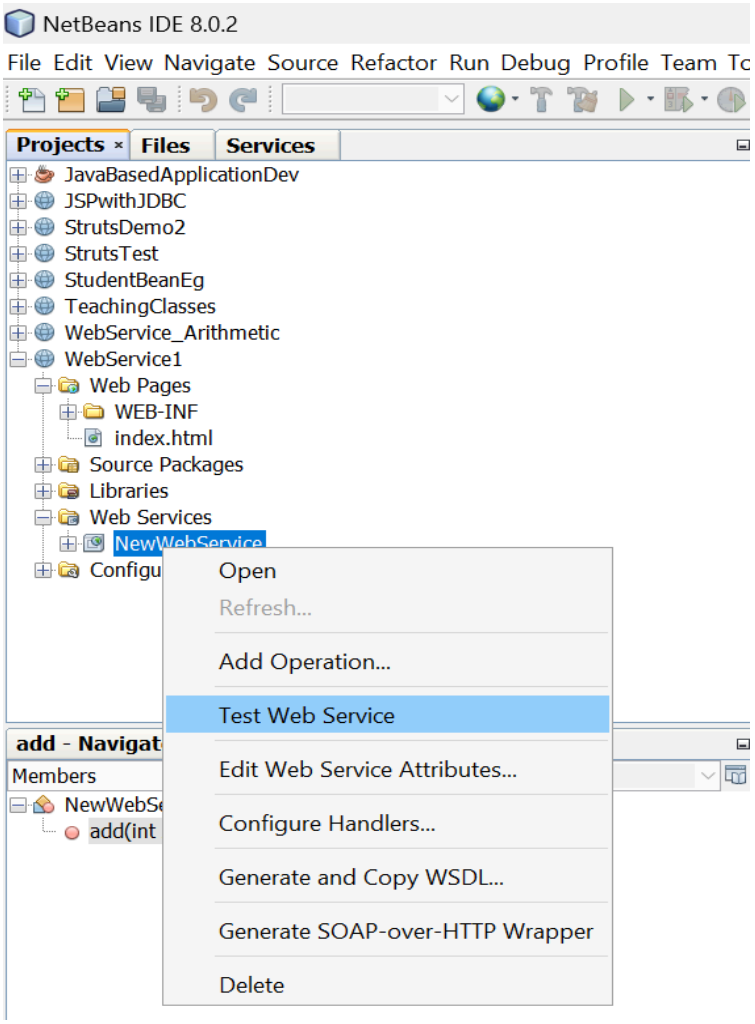


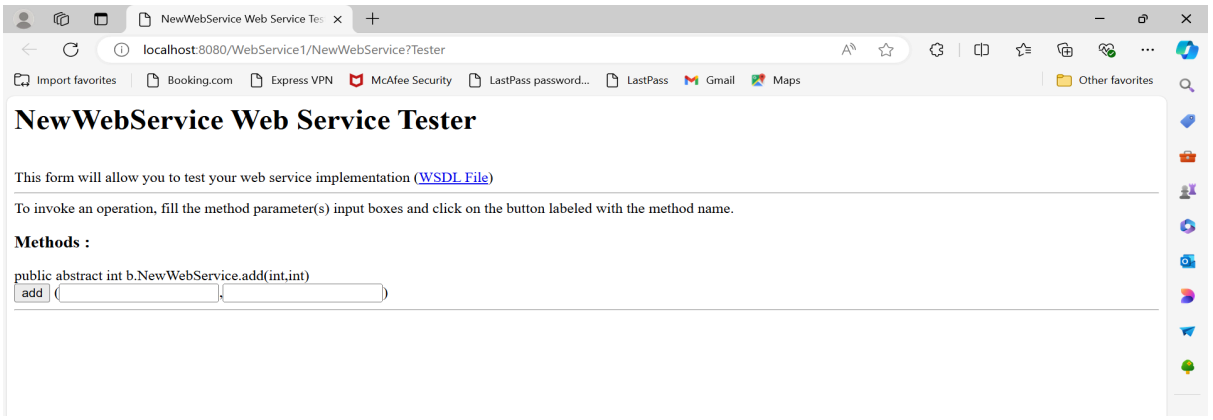
6. Now right click on project name and click on Deploy to deploy your project.

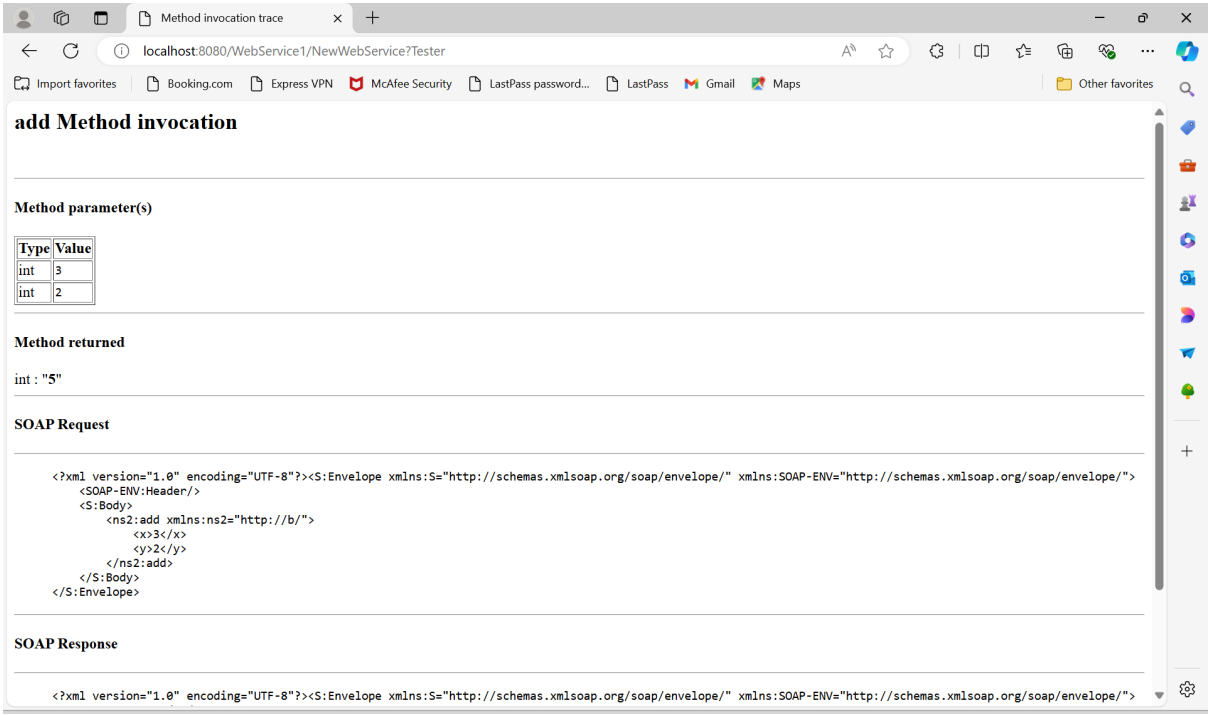


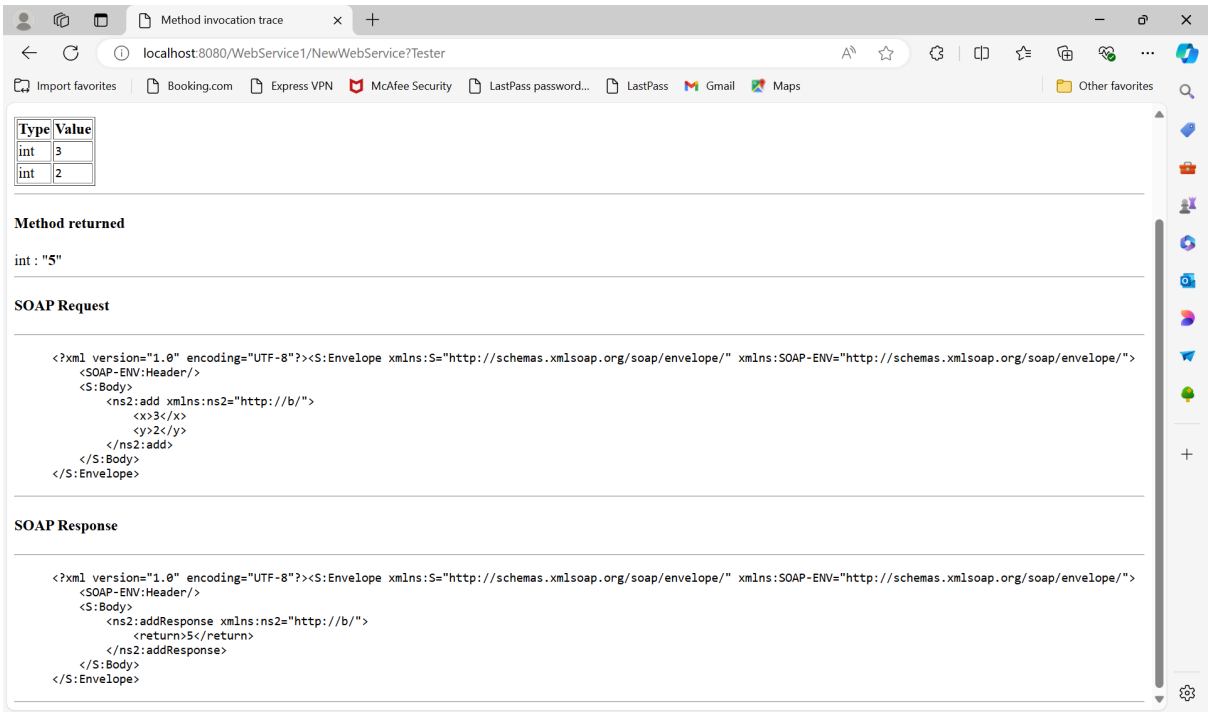


7. To test your web service follow the following process as in picture.

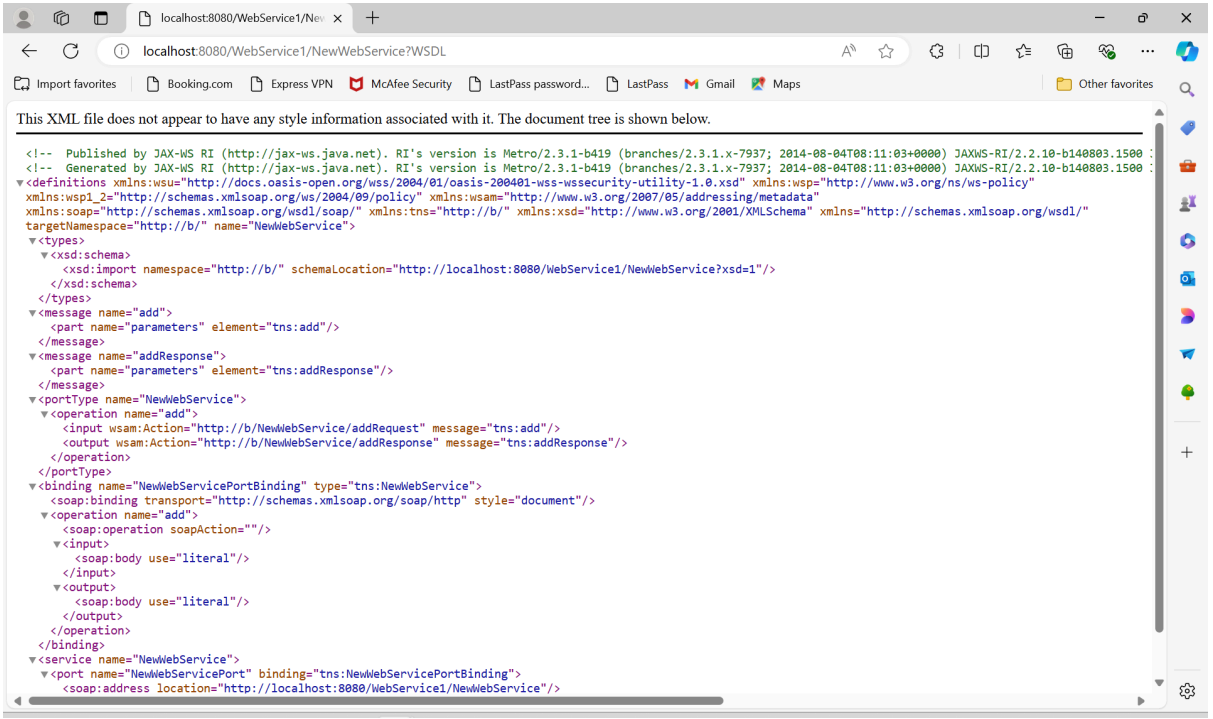

NetBeans IDE 8.0.2
File Edit View Navigate Source Refactor Run Debug Profile Team Tc
Projects
Files
Services
JavaBasedApplicationDev
JSPwithJDBC
StrutsDemo2
StrutsTest
StudentBeanEg
TeachingClasses
WebService_Arithmetic
WebService1
Web Pages
WEB-INF
index.html
Source Packages
Libraries
Web Services
NewWebService
Configu
Open
Refresh...
Add Operation...
Test Web Service
Edit Web Service Attributes...
Configure Handlers...
Generate and Copy WSDL...
Generate SOAP-over-HTTP Wrapper
Delete
add - Navigat
Members
NewWebSe
add(int


NewWebService Web Service Te
localhost:8080/WebService1/NewWebService?Tester
Import favorites
Booking.com
Express VPN
McAfee Security
LastPass password...
LastPass
Gmail
Maps
Other favorites
NewWebService Web Service Tester
This form will allow you to test your web service implementation (WSDL File)
To invoke an operation, fill the method parameter(s) input boxes and click on the button labeled with the method name.
Methods :
public abstract int b.NewWebService.add(int,int)
add


Method invocation trace
localhost:8080/WebService1/NewWebService?Tester
Import favorites
Booking.com
Express VPN
McAfee Security
LastPass password...
LastPass
Gmail
Maps
Other favorites
add Method invocation
Method parameter(s)
Type Value
int 3
int 2
Method returned
int : "5"
SOAP Request
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
<SOAP-ENV:Header/>
<S:Body>
<ns2:add xmlns:ns2="http://b/">
<x>3</x>
<y>2</y>
</ns2:add>
</S:Body>
</S:Envelope>
SOAP Response
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">


Method invocation trace
localhost:8080/WebService1/NewWebService?Tester
Import favorites
Booking.com
Express VPN
McAfee Security
LastPass password...
LastPass
Gmail
Maps
Other favorites
Type Value
int 3
int 2
Method returned
int : "5"
SOAP Request
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
<SOAP-ENV:Header/>
<S:Body>
<ns2:add xmlns:ns2="http://b/">
<x>3</x>
<y>2</y>
</ns2:add>
</S:Body>
</S:Envelope>
SOAP Response
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
<SOAP-ENV:Header/>
<S:Body>
<ns2:addResponse xmlns:ns2="http://b/">
<return>5</return>
</ns2:addResponse>
</S:Body>
</S:Envelope>

WSDL file:

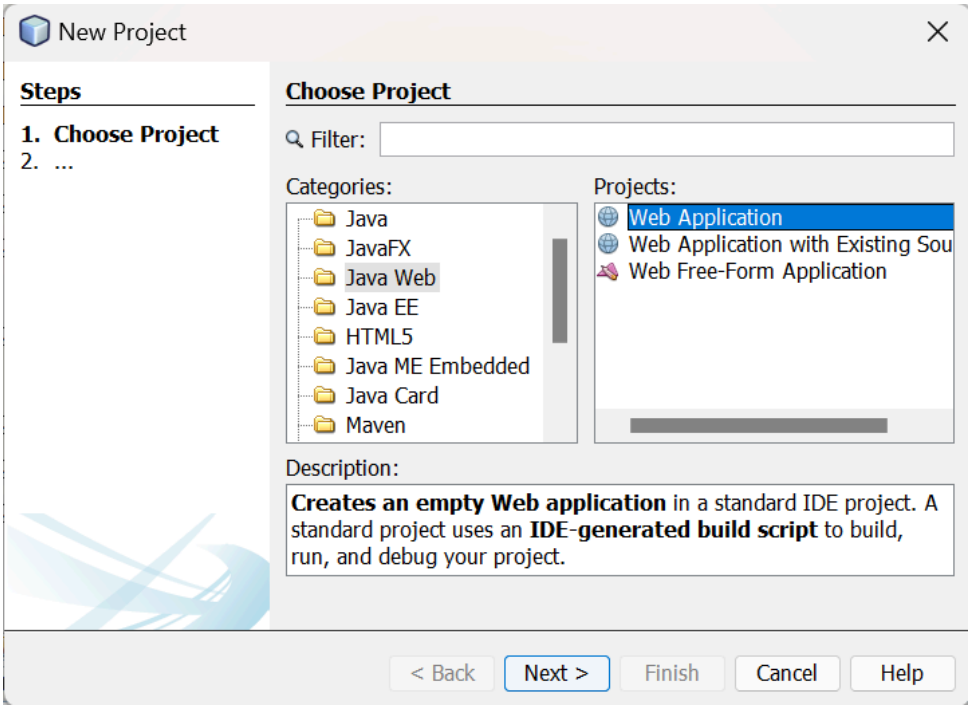


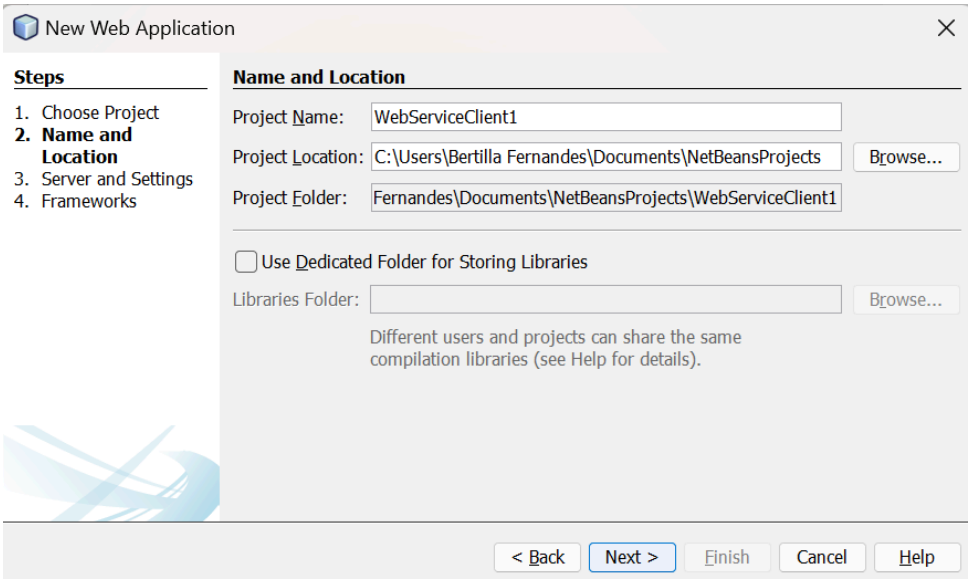
B.Create the Web Service Client

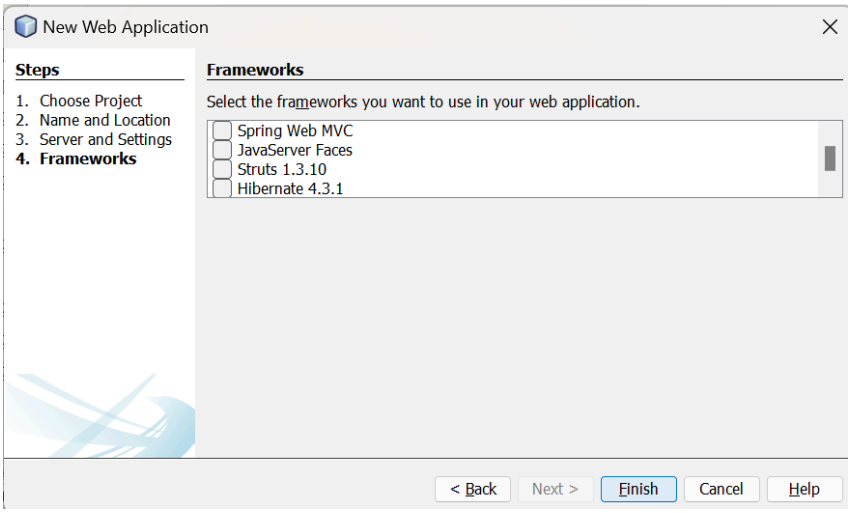
8.Create the Web Service client

Now to consume this web service we are creating a client.

Now create a new web application project with name as “WebServiceClient1”

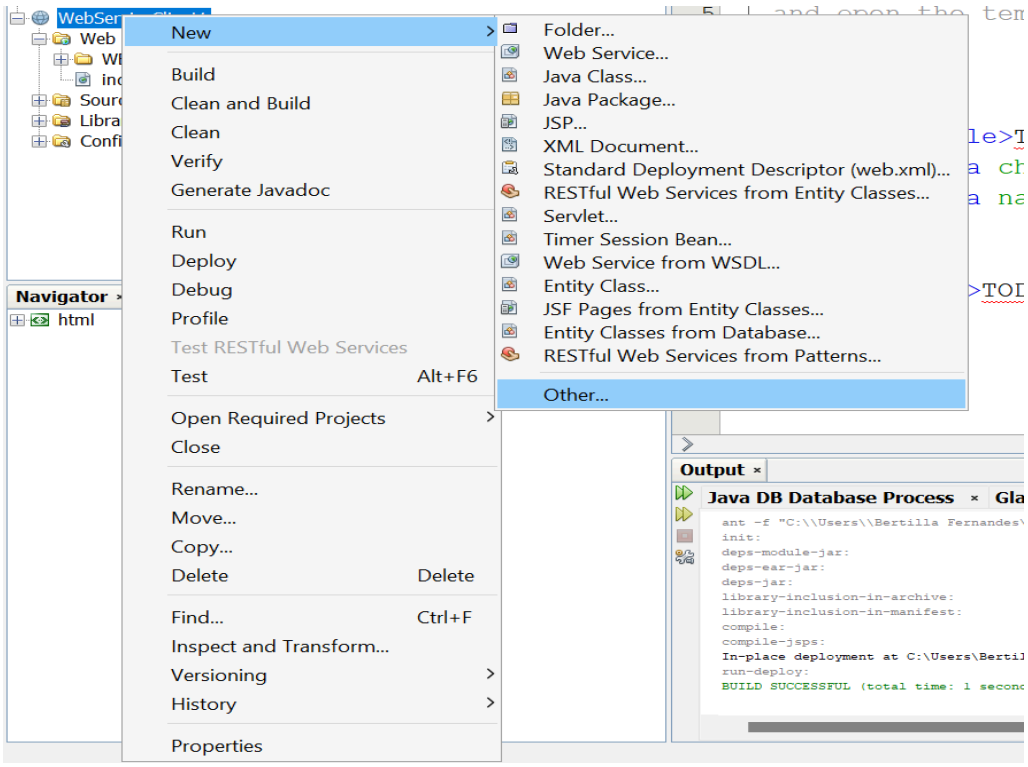


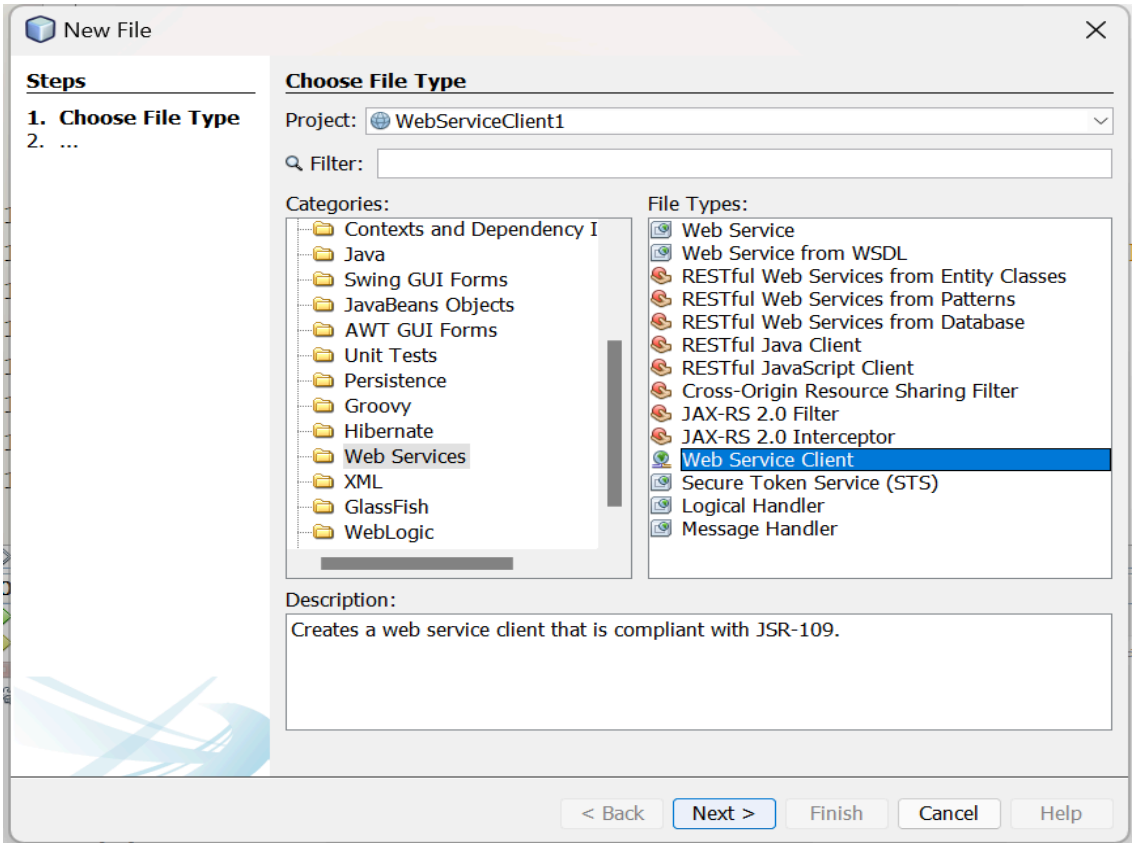




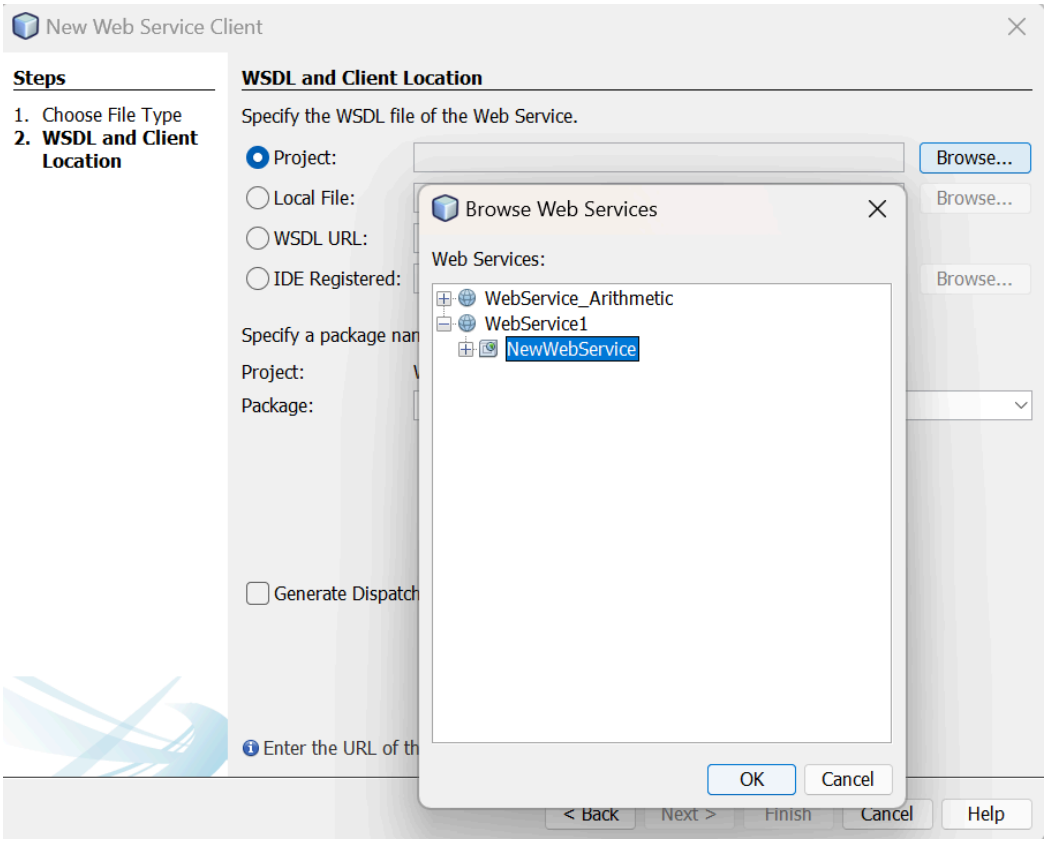
9. WebClient needs to access the WebService.

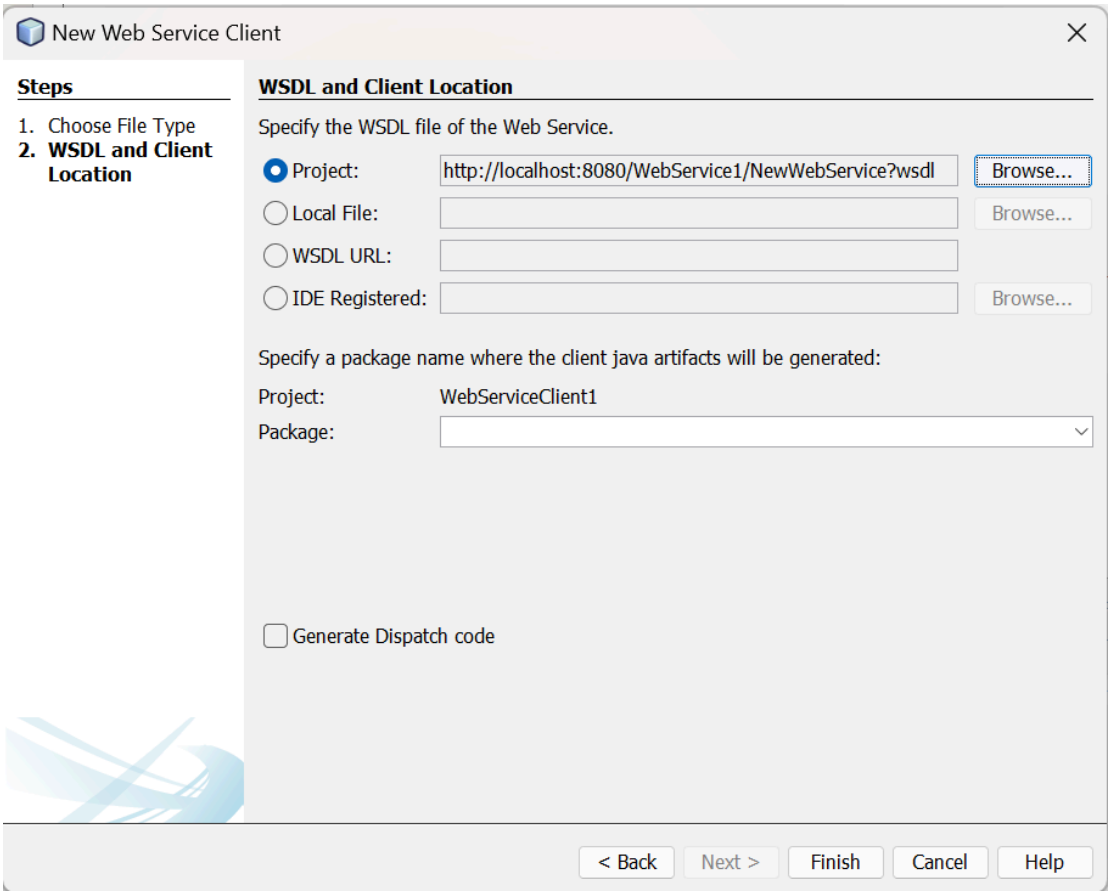
Right click on project and select Web Service Client





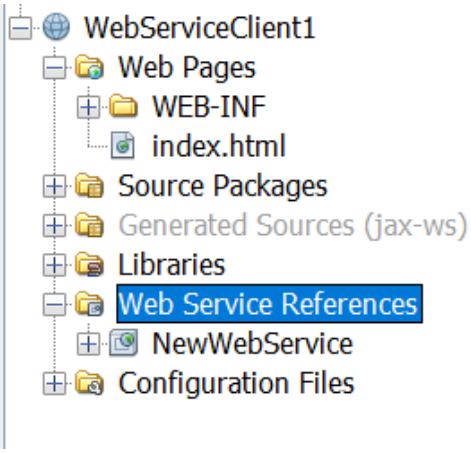
10. Provide the Project details by clicking on Browse as shown in the picture.





Specify the Project and Click on Finish.

Web Service References is added.

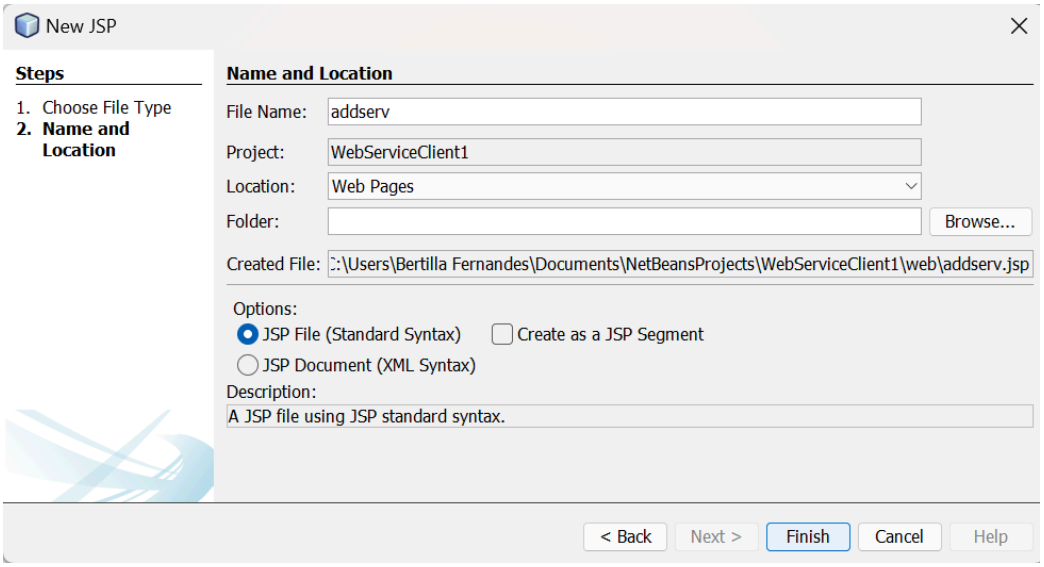


11. Create the index.html file and add the following code:

```
<html>
    <head>
        <title>TODO supply a title</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <form action="addserv.jsp">
            <input type="text" name="t1" value="">
            <input type="text" name="t2" value="">
            <input type ="submit" value="ADDSERVICE">
        </form>
    </body>
</html>
```

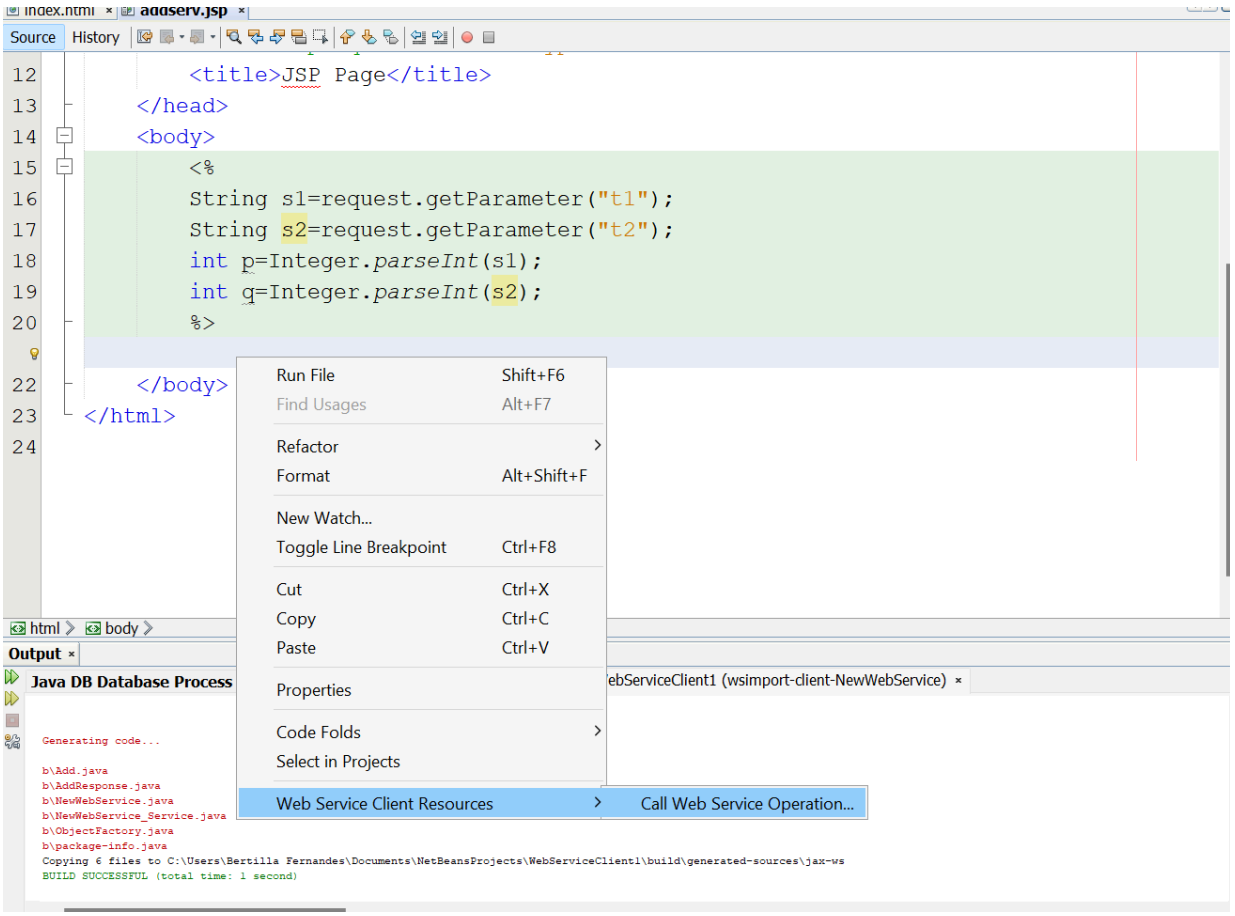
12. Create the JSP file:

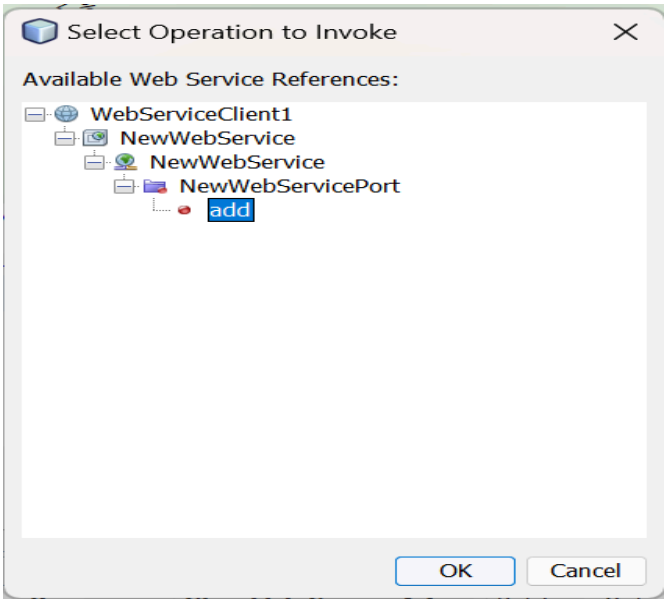
addserv.jsp



13.The JSP/Servlet file will need to call the webservice:

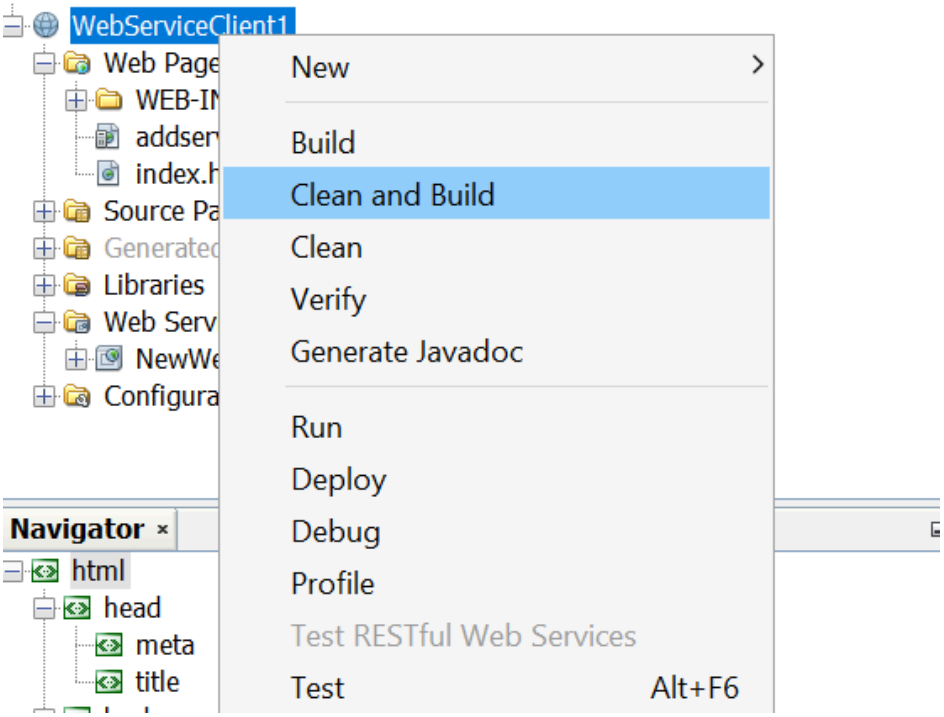
Right click in between the body section and select Call Web Service Operation as below.



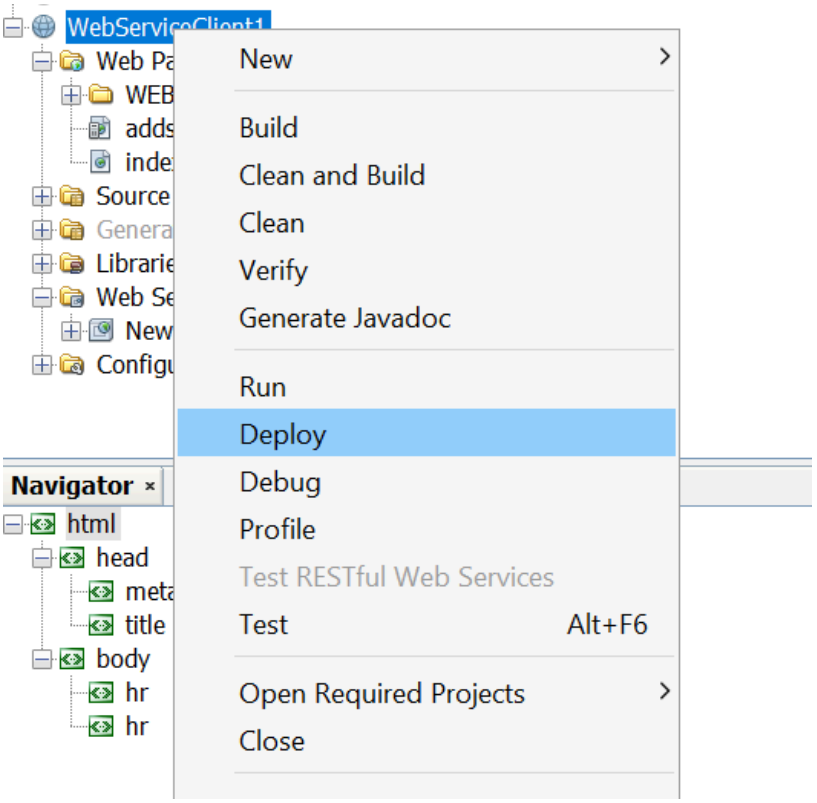




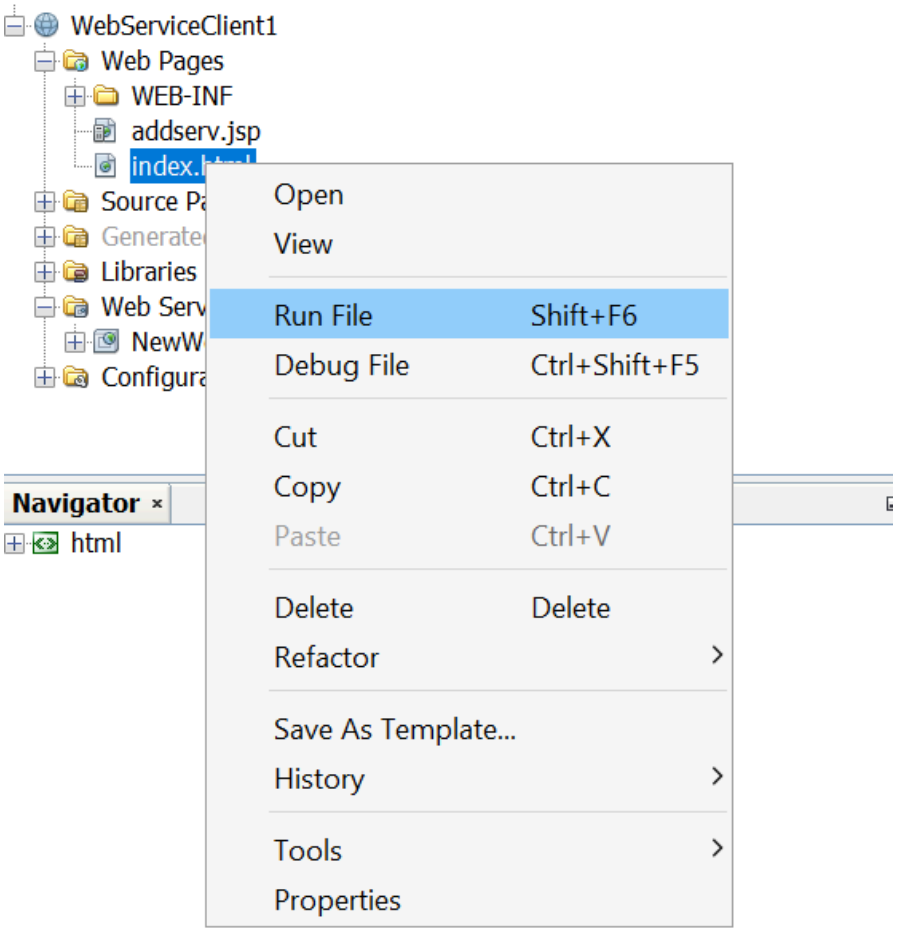
14. Now right click on project name and click on “Clean and Build” to clean and build your project.



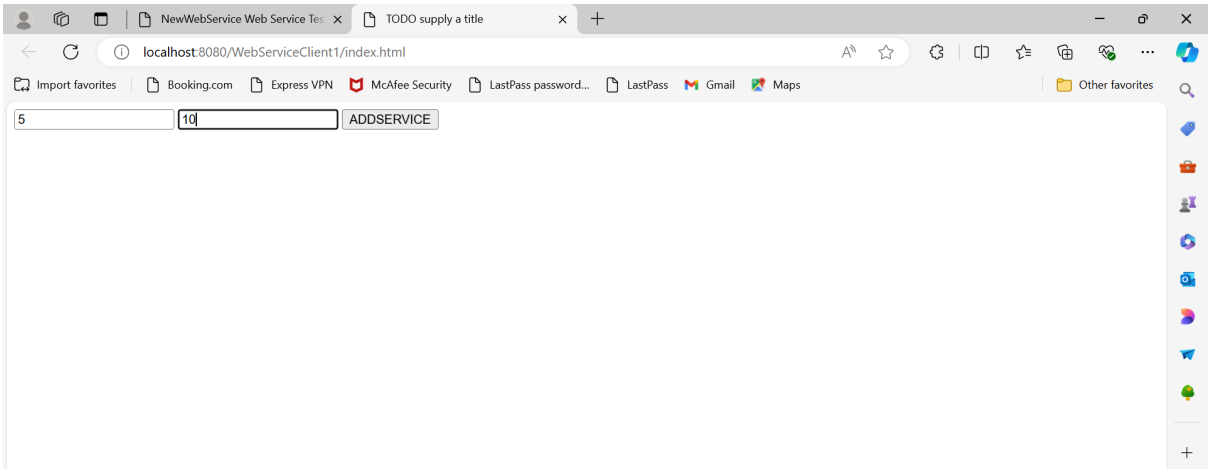
15. Now right click on project name and click on “Deploy” to deploy your project.Deploy:

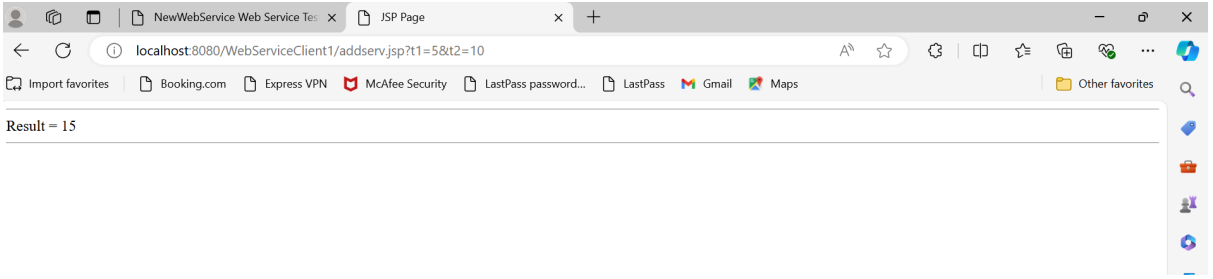


16. Run the index.html file


WebServiceClient1
Web Pages
WEB-INF
addserv.jsp
Source Pa
Generate
Libraries
Web Serv
NewW
Configura
Navigator
html
Open
View
Run File Shift+F6
Debug File Ctrl+Shift+F5
Cut Ctrl+X
Copy Ctrl+C
Paste Ctrl+V
Delete Delete
Refactor
Save As Template...
History
Tools
Properties

Output:


NewWebService Web Service Tes
TODO supply a title
localhost:8080/WebServiceClient1/index.html
Import favorites
Booking.com
Express VPN
McAfee Security
LastPass password...
LastPass
Gmail
Maps
Other favorites
5
10
ADDSERVICE


NewWebService Web Service Tes
JSP Page
localhost:8080/WebServiceClient1/addserv.jsp?t1=5&t2=10
Import favorites
Booking.com
Express VPN
McAfee Security
LastPass password...
LastPass
Gmail
Maps
Other favorites
Result = 15